E-Commerce Sales Data Analysis Project

Project Objective:

The objective of this project is to analyze e-commerce sales data in Excel to understand sales trends, customer behavior, and payment patterns using descriptive statistics and visualizations.

Dataset Description:

This dataset has 500 rows and 8 columns. It contains details about e-commerce sales such as orders, customers, products, prices, discounts, payment methods, and total amount.

Column Description

Column Name	Description
CUSTOMER ID	It represents customer id in each transaction.
STORE ID	It represents Store ID in each transaction.
PRODUCT ID	It represents Product ID in each Transaction.
CATEGORY	It represents Category of the Product
SUB CATEGORY	It represents the Type of the Product
BRAND	It represents the Brands of the Product
REGION	It represents the Region of sales
STORE TYPE	It represents the what type of store
PAYMENT TYPE	It represents type of the payment
STORE CITY	It represents the location of the store
UNIT PRICE	It represents Price of the Product with Discount
Total Amount	It represents Total Price after Discount

Visualization and Insights Arrived:

- Descriptive analysis:

  Various charts such as Bar charts, Column charts, Pie charts, and Sparklines were created in Excel to visualize sales distribution by payment type, store-wise sales, and product performance.

- Diagnostic Analysis:

  Descriptive statistics such as Mean, Minimum, Maximum, and Standard Deviation were calculated using Excel Analysis ToolPak. The average total sales value is approximately 0.00, which represents the typical transaction value

- Predictive Analysis:

  Based on historical sales trends, future sales are expected to remain stable with potential growth if effective discount strategies and popular payment methods are promoted.

- Prescriptive Analysis:

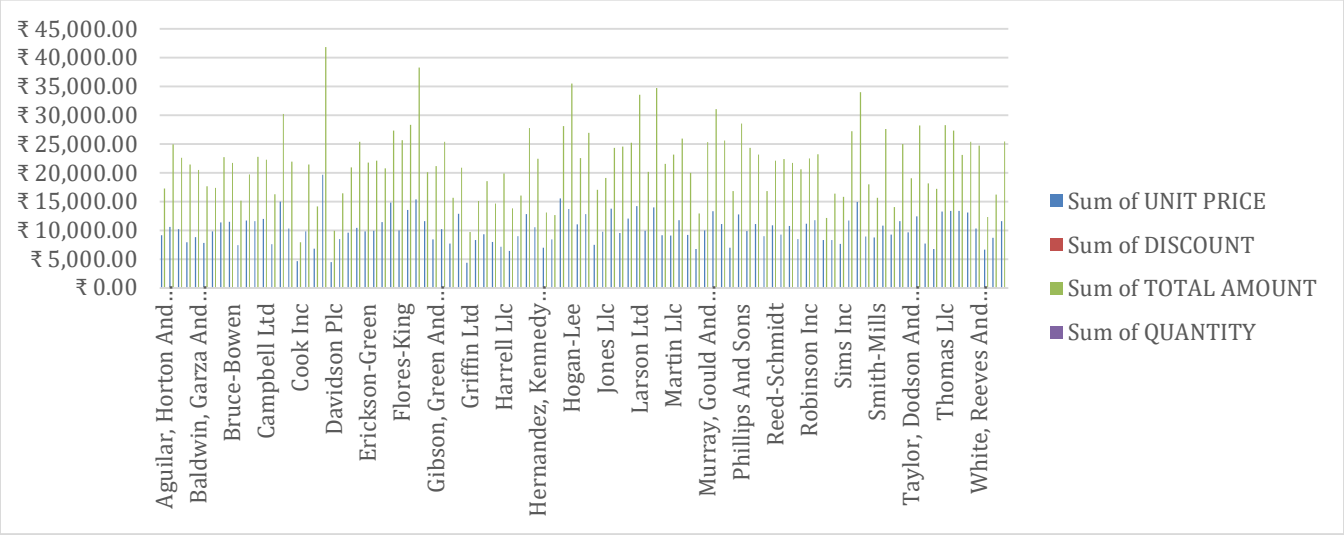
  The business should focus on optimizing product pricing, managing discounts effectively, and encouraging digital payment methods to enhance overall sales performance.

Insights & Findings

The analysis revealed that certain product categories and store locations contributed significantly to overall sales. The dashboard helped identify high-performing areas and opportunities for improvement. The use of interactive visuals made trend analysis efficient and user-friendly.

Conclusion:

The analysis concludes that the e-commerce platform has generated a total sales value of 0.00. Excel dashboards and charts provide valuable insights for data-driven decisions.

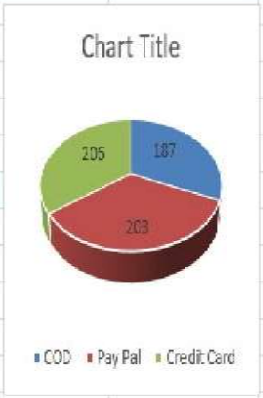


REGION	AVERAGE SALES
NORTH	₹ 1,086.73
EAST	₹ 1,087.20
WEST	₹ 1,087.76
SOUTH	₹ 1,087.49



REGION	TOTAL SALES
NORTH	₹ 21,66,944.98
EAST	₹ 21,64,620.40
WEST	₹ 21,70,088.49
SOUTH	₹ 21,68,447.39

PAYMENT	COUNTS
COD	187
Pay Pal	203
Credit Card	206





YEAR	TOTAL SALES DONE BY MALES
2023	2127273.555
2024	2162881.939
2025	2141980.042

YEAR	TOTAL SALES DONE BY MALES
2023	2146491.664
2024	2127781.636
2025	2119985.313



